

Active buzzer

DO

SOUND

What does it do?

An active buzzer makes one built-in tone. Switch it on and off for warning beeps or rhythms; its pitch stays the same.

Recognise it: check the packet for **active** or **built-in oscillator**. Shape, stickers and + marks do not prove the type.

Use a labelled **3.3 V GPIO-compatible** buzzer.

Uses

Smoke alarm



One warning tone is enough; no changing notes needed.

DETECT
Measure
smoke level

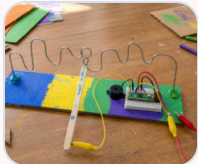


DECIDE
Is smoke
above the
limit?



DO
Repeat one
warning tone

Buzz wire game



Choose active: one fixed buzz clearly signals failure.

DETECT
Loop touches
wire



DECIDE
Is this the
third strike?



DO
Buzz at one
pitch



+ leg
Marked +, to
GPIO

- leg
Other leg, to
GND

Vent
Sound comes
out

Code

● Raspberry Pi

```
from gpiozero import Buzzer  
  
buzzer = Buzzer(12)  
  
buzzer.on()
```

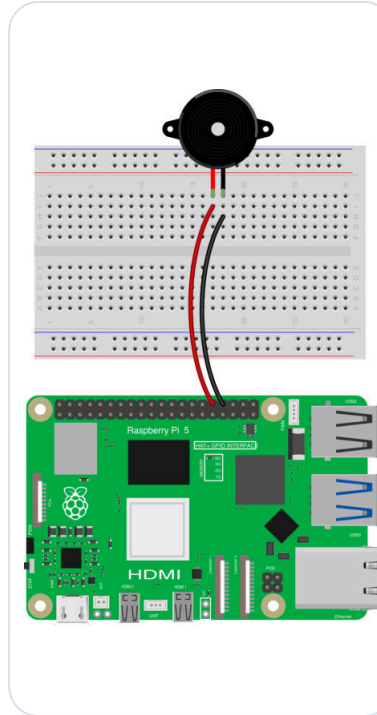
● Raspberry Pi Pico

```
from picozero import Buzzer  
  
buzzer = Buzzer(16)  
  
buzzer.on()
```

Also `buzzer.off()`, `buzzer.toggle()` and `buzzer.beep()`. Give `beep()` an `on_time` and an `off_time` to set the rhythm. Faster beeps do not mean a higher pitch.

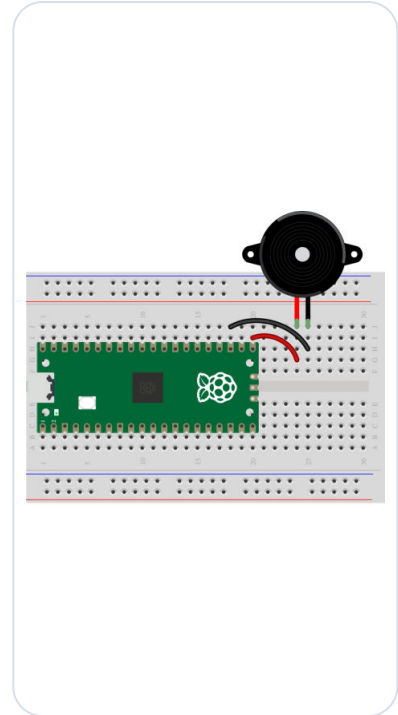
Wiring

RASPBERRY PI 5



GPIO 12 → + buzzer - → GND

PICO



GP16 → + buzzer - → GND